FNCE30007 Tutorial 8 Dividends

Question 1

Consider a stock index currently standing at 250. The dividend yield on the index is 4% per annum, and the risk-free rate is 6% per annum. A three-month European call option on the index with a strike price of 245 is currently worth 10. What is the value of a three-month put option on the index with a strike price of 245, if each index point is valued at $25?

Question 2

An index currently stands at 696 and has a volatility of 30% per annum. The risk-free rate of interest is 7% per annum and the index provides a dividend yield of 4% per annum. Calculate the value of a three-month European put with an exercise price of 700, if each index point is valued at $25.

Question 3

Suppose that the spot price of the Canadian dollar is U.S. $0.95 and that the Canadian dollar/U.S. dollar exchange rate has a volatility of 8% per annum. The risk-free rates of interest in Canada and the United States are 4% and 5% per annum, respectively. Calculate the value of a European call option to buy one Canadian dollar for U.S. $0.95 in nine months. Use put-call parity to calculate the price of a European put option to sell one Canadian dollar for U.S. $0.95 in nine months. What is the price of a call option to buy U.S. $0.95 with one Canadian dollar in nine months?

Question 4

Suppose that a portfolio is worth $60 million and the S&P 500 is at 1200. If the value of the portfolio mirrors the value of the index, what options should be purchased to provide protection against the value of the portfolio falling below $54 million in one year's time, assuming each option contract is for $100 times the index?

Question 5

Calculate u, d, and p when a binomial tree is constructed to value an option on a foreign currency. The tree step size is one month, the domestic interest rate is 5% per annum, the foreign interest rate is 8% per annum, and the volatility is 12% per annum.

Question 6

Consider an option on a dividend-paying stock when the stock price is $30, the exercise price is $29, the risk-free interest rate is 5% per annum, the volatility is 25% per annum, and the time to maturity is four months. The stock is due to go ex-dividend in 1.5 months and the expected dividend is 50 cents.

a. What is the price of the option if it is a European call? Check your answers using the simulator
b. What is the price of the option if it is a European put? Check your answers using the simulator
c. What happens to the price if the stock is expected to pay another 50 cents dividend in 6 months (ex-dividend date in 6 months)? Check your answers using the simulator. Comment on the findings.

Question 7

After reading the Virgin case, complete the following questions.

Part A - Can One Anchor Volatility Price Other Calls?

Use the implied volatility obtained from the 101C anchor option as the common σ for the calls below.

Strike K	Market Mid-Price	Common σ	BSM Value	Difference	Abs. % Error
95	10.080				
100	7.495				
101	7.065		7.065	0.000	0.00%
105	5.600				
110	4.215				

(a) Calculate the BSM values using the same σ for all five calls.

(b) Compare the BSM values with the market mid-prices. Does one constant volatility reproduce all call prices exactly?

(c) What does this suggest about the constant-volatility assumption?

Part B - Call Smile and Put Smile

Now infer an individual implied volatility from each observed option mid-price. Use the call prices to construct a call smile and the put prices to construct a put smile. Both curves use the same strikes.

7.1 Call Implied-Volatility Smile

Strike K	Call Mid-Price	Call Implied Volatility
80	21.800	
85	17.405	
90	13.435	
95	10.080	
100	7.495	
101	7.065	
105	5.600	
110	4.215	
115	3.185	
120	2.420	
125	1.840	
130	1.425	
135	1.110	
140	0.880	

7.2 Put Implied-Volatility Smile

Strike K	Put Mid-Price	Put Implied Volatility
80	0.680	
85	1.285	
90	2.310	

95	3.970	
100	6.375	
101	6.945	
105	9.475	
110	13.100	
115	17.075	
120	21.300	
125	25.730	
130	30.310	
135	35.000	
140	39.760	

Question: What do both curves suggest about the BSM assumption that volatility is constant across strikes?